

Bachelor in Computer Applications

Course Title: **Data Structure and Algorithms**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Explain algorithm to convert an infix expression to postfix using stack? Use this algorithm to convert $(A+B)*C-D$ to postfix.
2. Explain merge sort along with its time complexity. Use this algorithm to sort array of numbers given below: 25, 37, 48, 25, 23, 17, 31, 45, 7, 21, 15, 8, 11
3. What is shortest path algorithm? Use Dijkstra's algorithm to find shortest path between the vertices of a and z in the graph given below.
4. What is data Structure? Explain an array as an abstract data type.
5. Explain big oh(O) notation with suitable example.
6. Define priority queue. How do you implement priority queue? Explain.
7. Define recursion. Explain Tower of Hanoi algorithm in detail.
8. How can you implement queue using linked list? Explain.
9. What is binary tree? Explain different application of binary tree.
10. Explain sequential search. How is it different from binary search?
11. Define hashing. Explain quadratic probing with example.
12. What is graph traversal? Explain breadth first search.